ICT ACADEMY HANDS-ON TRAINING ON VLSI LAYOUT DESIGN

STUDENT DETAILS

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- Department: B.E. ECE – III Year

Day 1: VLSI Design Flow and Fabrication Foundations

- Evolution of VLSI: Studied the progression from SSI (1960s) to Giga-Scale Integration involving billions of transistors.
- Design Cycle: Explored VLSI flow from System Specification and Architectural Design to RTL Coding (Verilog/VHDL).
- Physical Design Steps: Partitioning, Floorplanning, Placement, Routing, and Physical Verification.
- Fabrication Process: Wafer Preparation, Oxidation, Photolithography, Etching, Ion Implantation, Polysilicon Gate Formation, Metallization, and CMP.

Day 2: Introduction to Cadence Virtuoso – CMOS Inverter

- Environment Setup: Initialized Virtuoso tool, created libraries, and attached technology files (gpdk 90nm/180nm).
- Schematic Entry: Designed CMOS inverter using PMOS and NMOS transistors.
- Symbol Creation: Generated reusable symbols for hierarchical design.
- Layout Basics: Studied layers such as N-well, Active, Poly, and Metal1.

Day 3: NAND Gate Design and Logic Verification

- Circuit Implementation: Designed 2-input CMOS NAND gate (Parallel PMOS, Series NMOS).
- Simulation: Performed transient analysis to verify NAND truth table.
- Layout Strategy: Applied diffusion sharing to reduce parasitic capacitance and area.

- Verification: Performed DRC (Design Rule Check) for manufacturability.

Day 4: Complex Logic Gates (NOR and EXOR)

- NOR Gate Design: Implemented 2-input NOR gate (Series PMOS, Parallel NMOS).
- EXOR Implementation: Developed EXOR using static CMOS and transmission gates.
- Optimization: Compared area-efficiency trade-offs of different logic structures.
- Routing: Practiced metal routing and via placement.

Day 5: Advanced Analog Layout & Reliability Guidelines

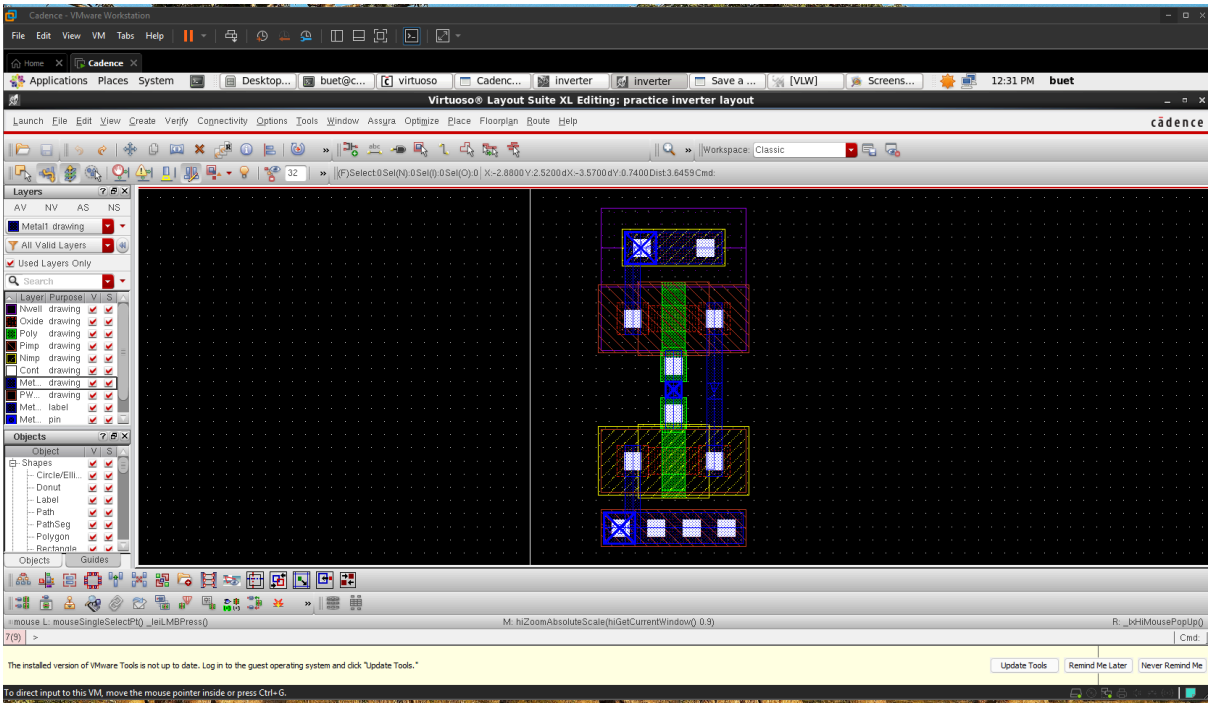
- Layout Guidelines: Studied matching techniques, floorplanning, and routing symmetry.
- Latch-up Prevention: Used guard rings and tap cells.
- Electromigration: Prevented metal damage by increasing wire width.
- IR Drop: Managed voltage drops using higher metal layers and stacking.
- Antenna Effect: Mitigated charge accumulation during fabrication.
- Process Variations: Studied impact on sheet resistance (Rs) and threshold voltage (Vt).

Resources & References

Project Code: https://github.com/syedkit/cadence_virtuoso_layout.git

Video Documentation: https://youtu.be/A-orhMjjW_k?si=_EIsfJQIg-BxL0s5

IMAGES:-


Virtuoso® Layout Suite XL Editing: practice inverter layout
Layers
Objects

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GPS Map Camera
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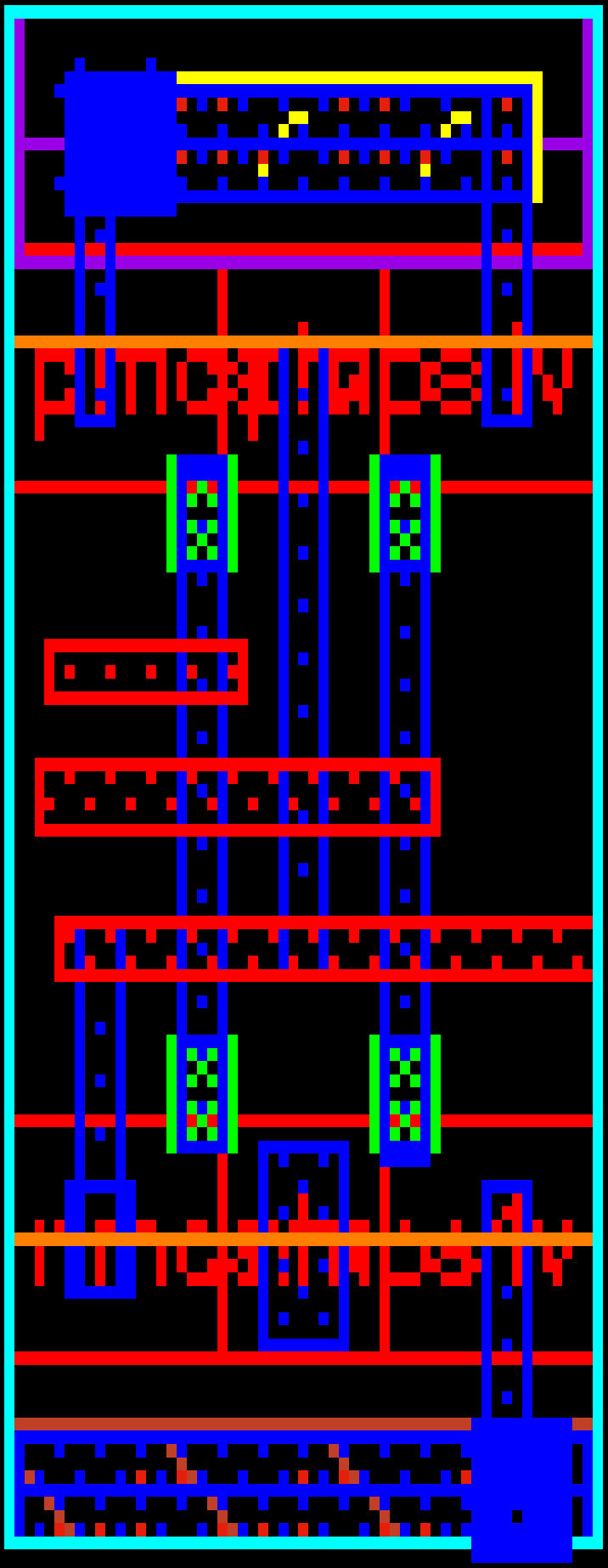

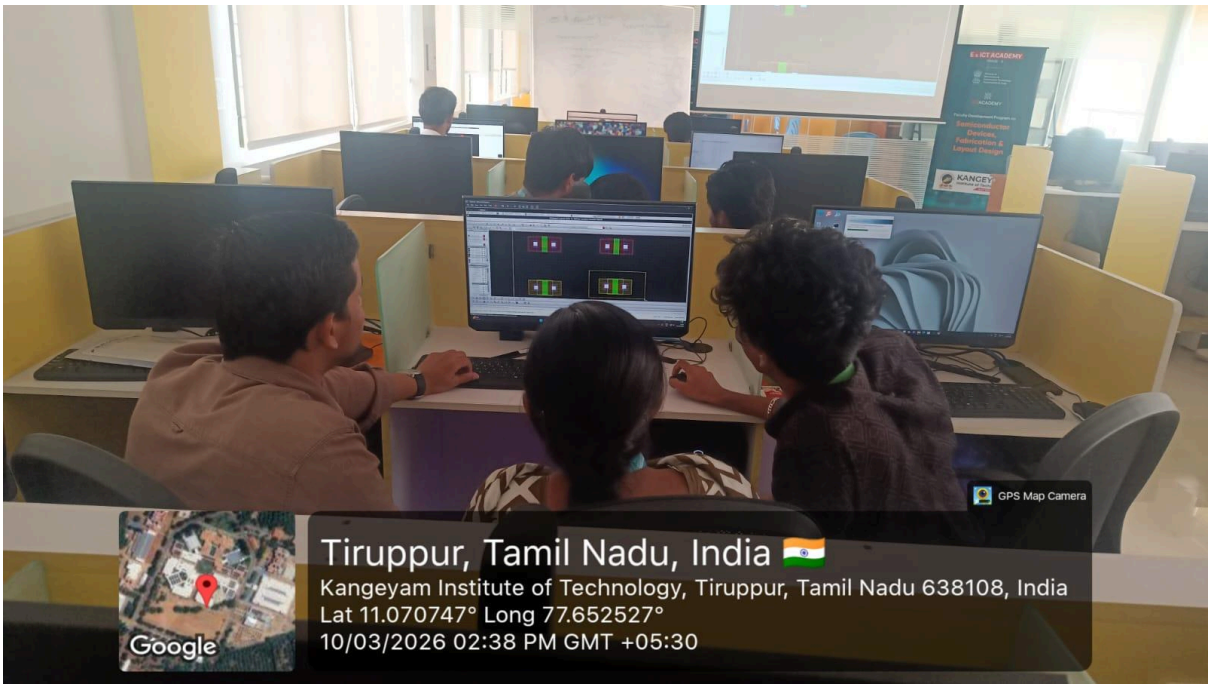
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